

Beyond Arbitrage: Hungary and the Reflexivity Supermodel

Lord Soumadeep Ghosh and Barbara Bukhvalova with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper proposes Reflexivity as the candidate C4Hypermodel supermodel for explaining the economic structure of Hungary, while treating Arbitrage as an important mechanism operating within the broader system. The central distinction is between a mechanism that exploits a price discrepancy and a system in which expectations, policy responses, market prices, capital flows, and observed fundamentals recursively influence one another.

The paper develops the proposition across economics, finance, econometrics, statistics, mathematical economics, physics, chemistry, biology, psychology, philosophy, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, computer science, algorithms, data science, and artificial intelligence.

The analysis builds on two complementary empirical architectures. The first examines bilateral and triangular currency arbitrage among nine nuclear nations using graph theory and exhaustive path enumeration. The second examines Scandinavian currencies using covered-interest-parity deviations, high-frequency foreign-exchange data, triangular arbitrage, transaction costs, and machine-learning prediction. These studies demonstrate that arbitrage and reflexivity should not be treated as synonymous.

The central hypothesis is therefore:

$$\boxed{\text{Hungary} \longrightarrow \text{Reflexivity}} \quad (1)$$

rather than

$$\text{Hungary} \longrightarrow \text{Arbitrage}. \quad (2)$$

The proposed Hungarian model is explicitly presented as a testable hypothesis rather than as an established empirical conclusion. Its purpose is to identify whether feedback between expectations, policy, financial markets, institutions, and economic outcomes provides greater explanatory power than a pure arbitrage model.

The paper ends with “The End”

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1 Introduction

The purpose of this paper is to investigate a deceptively simple question:

$$\text{Which C4Hypermodel supermodel best explains Hungary?} \quad (3)$$

The proposed answer is Reflexivity.

This answer should not be interpreted as a claim that Hungary is free from arbitrage opportunities. Nor should it be interpreted as a claim that arbitrage is economically unimportant. Rather, the proposition is that Arbitrage is insufficient as a primary explanatory model for the Hungarian system.

The distinction is fundamental.

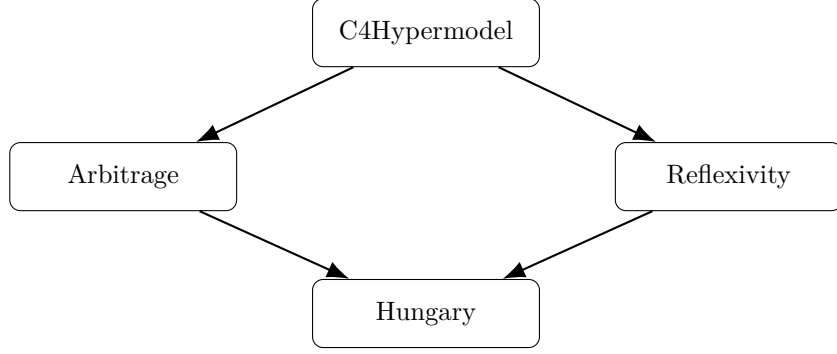


Figure 1: Architecture of Hungary

Arbitrage begins with a discrepancy:

$$\text{price discrepancy} \longrightarrow \text{trade} \longrightarrow \text{convergence.} \quad (4)$$

Reflexivity begins with an interaction between perception and reality:

$$\text{belief} \longrightarrow \text{action} \longrightarrow \text{market outcome} \longrightarrow \text{changed belief.} \quad (5)$$

The second structure is recursive.

The observer is not necessarily external to the system. The participant can affect the object being observed.

This distinction becomes especially important in financial systems because prices are simultaneously measurements and incentives. A price is an observation of market conditions, but it can also change behaviour.

The proposed Hungarian supermodel therefore treats economic agents as participants in a dynamically evolving system rather than as passive observers of an externally determined equilibrium.

2 The C4Hypermodel Perspective

The C4Hypermodel is treated in this paper as a higher-order architecture containing several supermodels.

One of these supermodels is Arbitrage.

The present paper proposes that another supermodel, Reflexivity, is more appropriate as the primary explanatory structure for Hungary.

The architecture can be represented as follows.

The important point is that the two supermodels need not be mutually exclusive.

A financial system may contain arbitrage opportunities while its broader dynamics remain reflexive.

3 Arbitrage as a Supermodel

3.1 Bilateral Arbitrage

Let $S_{A/B}$ denote the spot exchange rate of currency A in terms of currency B .

A bilateral arbitrage opportunity exists when:

$$S_{A/B}S_{B/A} \neq 1. \quad (6)$$

Define the bilateral profit factor as:

$$\pi_2 = S_{A/B}S_{B/A} - 1. \quad (7)$$

If:

$$\pi_2 > 0, \quad (8)$$

a theoretical arbitrage opportunity exists before transaction costs and other execution constraints.

3.2 Triangular Arbitrage

For three currencies A , B , and C ,

$$\pi_3 = S_{A/B}S_{B/C}S_{C/A} - 1. \quad (9)$$

A positive value indicates a theoretical profitable cycle.

The corresponding no-arbitrage condition is:

$$S_{A/B}S_{B/C}S_{C/A} = 1. \quad (10)$$

The nine-nation currency study tested 36 bilateral pairs and 504 directed triangular paths and reported no exploitable arbitrage under its January 2026 market snapshot. The study also explicitly distinguished theoretical arbitrage from practical accessibility in the North Korean dual-rate system.

4 Currency Networks and Negative Cycles

A currency system can be represented as a directed graph:

$$G = (V, E, w). \quad (11)$$

The vertices are currencies, the directed edges are exchange-rate relationships, and the edge weight can be defined as:

$$w(u, v) = -\log S_{u/v}. \quad (12)$$

For a closed cycle C ,

$$C = (v_1, v_2, \dots, v_k, v_1), \quad (13)$$

the cycle corresponds to an arbitrage opportunity when:

$$\sum_{i=1}^k w(v_i, v_{i+1}) < 0. \quad (14)$$

This transforms multiplicative exchange-rate relationships into an additive graph problem.

The corresponding arbitrage factor is:

$$\pi_C = \prod_{i=1}^k S_{v_i/v_{i+1}} - 1. \quad (15)$$

Taking logarithms gives:

$$\log(1 + \pi_C) = \sum_{i=1}^k \log S_{v_i/v_{i+1}} \quad (16)$$

$$= -\sum_{i=1}^k w(v_i, v_{i+1}). \quad (17)$$

Consequently:

$$\pi_C > 0 \iff \sum_{i=1}^k w(v_i, v_{i+1}) < 0. \quad (18)$$

Bellman–Ford therefore provides a natural computational method for negative-cycle detection.

5 Why Arbitrage Is Not Reflexivity

Arbitrage is fundamentally a convergence mechanism.

Suppose:

$$P_A > P_B. \quad (19)$$

An arbitrageur purchases the cheaper asset and sells the more expensive asset.

Under idealized assumptions:

$$P_A - P_B \longrightarrow 0. \quad (20)$$

The arbitrageur acts as a force toward consistency.

Reflexivity can behave differently.

Suppose an agent expects:

$$E_t[P_{t+1}] > P_t. \quad (21)$$

The agent buys the asset.

Aggregate buying pressure changes the price:

$$Q_t^d \uparrow \implies P_t \uparrow. \quad (22)$$

The increase in price may then reinforce the original expectation:

$$P_t \uparrow \implies E_t[P_{t+1}] \uparrow. \quad (23)$$

Thus:

$$E_t \longrightarrow Q_t \longrightarrow P_t \longrightarrow E_{t+1}. \quad (24)$$

This is a feedback loop rather than a simple convergence mechanism.

6 A Reflexive Economic System

Let the state of the economic system at time t be:

$$X_t = \begin{bmatrix} P_t \\ E_t \\ K_t \\ I_t \\ Y_t \\ \pi_t \\ R_t \end{bmatrix}, \quad (25)$$

where P_t denotes market prices, E_t expectations, K_t capital flows, I_t institutional conditions, Y_t economic activity, π_t inflation, and R_t relevant policy variables.

A general nonlinear dynamic system is:

$$X_{t+1} = F(X_t, u_t, \varepsilon_t). \quad (26)$$

Here u_t represents policy and strategic actions and ε_t represents shocks.

A reflexive system additionally permits expectations to depend upon anticipated future states:

$$E_t = h(X_t, E_{t+1}^e). \quad (27)$$

The system therefore contains an endogenous information-feedback channel.

7 The Hungarian Hypothesis

The proposed supermodel is:

$$\boxed{\mathcal{M}_{HU} = \mathcal{R}} \quad (28)$$

where $\mathcal{R}$ denotes Reflexivity.
The hypothesis is:

$$H_0 : \mathcal{M}_{HU} = \mathcal{A}, \quad (29)$$

where $\mathcal{A}$ denotes Arbitrage, against:

$$H_1 : \mathcal{M}_{HU} = \mathcal{R}. \quad (30)$$

However, this is not a conventional mutually exclusive statistical test, because Arbitrage may remain a mechanism within a Reflexive system.

A more appropriate decomposition is:

$$\mathcal{M}_{HU} = \mathcal{R} + \mathcal{A} + \mathcal{M} + \mathcal{P}, \quad (31)$$

where:

$$\mathcal{R} = \text{Reflexive feedback}, \quad (32)$$

$$\mathcal{A} = \text{Arbitrage}, \quad (33)$$

$$\mathcal{M} = \text{Market microstructure}, \quad (34)$$

$$\mathcal{P} = \text{Policy and institutional response}. \quad (35)$$

The empirical question becomes one of explanatory dominance rather than simple exclusion.

8 Why Hungary Is Interesting

Hungary provides a useful laboratory for studying the interaction of:

$$\{\text{currency, monetary policy, fiscal policy, expectations, European integration, capital flows}\}. \quad (36)$$

The relevant question is not whether the Hungarian forint can ever experience a price discrepancy. It is whether the observed dynamics are better characterized by:

$$\text{discrepancy} \rightarrow \text{arbitrage} \rightarrow \text{convergence}, \quad (37)$$

or by:

$$\text{expectations} \rightarrow \text{positioning} \rightarrow \text{prices} \rightarrow \text{policy} \rightarrow \text{expectations}. \quad (38)$$

The second mechanism is the candidate Reflexivity supermodel.

9 Covered Interest Parity

The Scandinavian study provides an appropriate methodological bridge between arbitrage and reflexivity. Covered interest parity can be expressed as:

$$(1 + i_A) \frac{F_{A/B}}{S_{A/B}} = 1 + i_B. \quad (39)$$

Define the CIP deviation as:

$$CIP_{A/B} = (1 + i_A) \frac{F_{A/B}}{S_{A/B}} - (1 + i_B). \quad (40)$$

The Hungarian empirical project should estimate:

$$CIP_{HUF/X,t} \quad (41)$$

for relevant currency pairs X .

The critical distinction is between an observed deviation and an exploitable arbitrage opportunity.

$$\text{Observed deviation} \neq \text{riskless profit}. \quad (42)$$

Transaction costs, liquidity, latency, market impact, settlement risk, regulatory constraints, and capital controls must be incorporated.

10 Transaction Costs

Let:

$$C_t = c_t^{\text{spread}} + c_t^{\text{commission}} + c_t^{\text{settlement}} + c_t^{\text{impact}} + c_t^{\text{latency}}. \quad (43)$$

Net arbitrage profit is:

$$\Pi_t^{\text{net}} = \Pi_t^{\text{gross}} - C_t. \quad (44)$$

An apparent arbitrage opportunity is economically actionable only if:

$$\Pi_t^{\text{net}} > 0. \quad (45)$$

The Scandinavian study reports this distinction explicitly by adjusting its arbitrage analysis for bid-ask spreads, latency, and execution slippage.

11 Reflexivity and Expectations

Let the representative market expectation be:

$$E_t[P_{t+1}]. \quad (46)$$

Let the demand function depend on expected return:

$$Q_t^d = D(E_t[P_{t+1}] - P_t). \quad (47)$$

Suppose price responds to excess demand:

$$P_{t+1} = P_t + \lambda(Q_t^d - Q_t^s). \quad (48)$$

Substitution produces:

$$P_{t+1} = P_t + \lambda[D(E_t[P_{t+1}] - P_t) - Q_t^s]. \quad (49)$$

The expectation therefore enters the price process.

If expectations themselves depend on prices,

$$E_{t+1}[P_{t+2}] = G(P_{t+1}, Z_{t+1}), \quad (50)$$

then the system is recursively coupled.

This is the mathematical core of the Reflexivity Supermodel.

12 A Stability Condition

Consider the simplified system:

$$P_{t+1} = a + bE_t[P_{t+1}]. \quad (51)$$

Let expectations satisfy:

$$E_t[P_{t+1}] = c + dP_t. \quad (52)$$

Then:

$$P_{t+1} = a + bc + bdP_t. \quad (53)$$

Define:

$$\phi = bd. \quad (54)$$

The fixed point is locally stable when:

$$|\phi| < 1. \quad (55)$$

It becomes unstable when:

$$|\phi| > 1. \quad (56)$$

Thus reflexivity need not imply instability.

It implies feedback.

The sign and magnitude of the feedback determine whether the system converges, oscillates, or becomes unstable.

13 Physics Analogy

The mathematical structure resembles a nonlinear dynamical system.

Let:

$$\frac{dX}{dt} = F(X). \quad (57)$$

At an equilibrium X^* ,

$$F(X^*) = 0. \quad (58)$$

A first-order approximation gives:

$$\frac{d\delta X}{dt} = J(X^*)\delta X, \quad (59)$$

where J is the Jacobian matrix.

The eigenvalues of J determine local stability.

In a reflexive economic system, the equivalent Jacobian can contain feedback terms from expectations, prices, policy, and capital flows.

$$J = \begin{bmatrix} \partial P / \partial P & \partial P / \partial E & \partial P / \partial K \\ \partial E / \partial P & \partial E / \partial E & \partial E / \partial K \\ \partial K / \partial P & \partial K / \partial E & \partial K / \partial K \end{bmatrix}. \quad (60)$$

The economic system can therefore be studied using tools from nonlinear dynamics.

14 Chemistry Analogy

Chemical reaction systems provide another useful analogy.

Suppose:



The forward and reverse reactions alter concentrations, and the altered concentrations change subsequent reaction rates.

Likewise, in a reflexive market:

$$\text{belief} \rightleftharpoons \text{market outcome}. \quad (62)$$

The analogy is structural rather than literal.

A market is not a chemical reaction.

The useful insight is that a state variable can simultaneously be an outcome of a process and an input into its subsequent evolution.

15 Biological Analogy

Biological systems also contain feedback.

Consider a simplified population system:

$$N_{t+1} = N_t + rN_t \left(1 - \frac{N_t}{K}\right). \quad (63)$$

The population changes the environment in which future population growth occurs.

Similarly:

$$\text{economic activity} \rightarrow \text{expectations} \rightarrow \text{capital allocation} \rightarrow \text{economic activity}. \quad (64)$$

The economic analogy is again structural.

Reflexivity therefore belongs naturally to the broader study of adaptive systems.

16 Psychology and Behavioral Economics

A reflexive model requires a theory of expectations.

Agents may update expectations according to:

$$E_{t+1} = E_t + \alpha (X_t - E_t), \quad (65)$$

where α represents the speed of updating.

A heterogeneous population may instead contain several updating rules:

$$E_{i,t+1} = f_i(X_t, E_{i,t}), \quad i = 1, \dots, N. \quad (66)$$

The resulting aggregate expectation is:

$$\bar{E}_t = \sum_{i=1}^N w_{i,t} E_{i,t}. \quad (67)$$

The weights $w_{i,t}$ may themselves depend on performance.

This creates evolutionary feedback:

$$\text{strategy performance} \rightarrow \text{strategy weight} \rightarrow \text{aggregate behaviour}. \quad (68)$$

The model therefore connects behavioural economics with adaptive expectations.

17 Psychiatry: A Boundary Condition

Psychiatric concepts should be used with particular caution in economic modelling.

The present paper does not diagnose individuals or populations.

Instead, it distinguishes:

$$\text{individual psychological process} \neq \text{clinical psychiatric diagnosis.} \quad (69)$$

The relevant economic concepts are collective expectations, attention, belief updating, uncertainty perception, and behavioural response.

No clinical inference about Hungarian citizens, policymakers, investors, or other groups follows from the Reflexivity Supermodel.

18 Philosophy of Knowledge

Reflexivity raises an epistemological problem.

Suppose:

$$M_t = f(D_t) \quad (70)$$

is a model constructed from data D_t .

If agents observe M_t and modify their behaviour because of it, then:

$$D_{t+1} = g(D_t, M_t). \quad (71)$$

Therefore:

$$M_t \rightarrow D_{t+1} \rightarrow M_{t+1}. \quad (72)$$

The model becomes part of the data-generating process.

This is one reason reflexive systems are epistemically difficult.

The distinction between observer and observed system becomes porous.

19 Political Science

Political institutions can be represented as strategic agents.

Let:

$$u_j = u_j(X_t, a_{j,t}, a_{-j,t}) \quad (73)$$

denote the utility of political actor j .

The actor chooses:

$$a_{j,t} = \arg \max_{a_j} u_j(X_t, a_j, a_{-j,t}). \quad (74)$$

Market participants anticipate these actions:

$$E_t[a_{j,t+1}] \neq \text{exogenous.} \quad (75)$$

Political decisions therefore become economic state variables when they affect expectations, taxation, spending, regulation, monetary policy, or external relations.

20 Geopolitics and International Relations

International relations introduces strategic externalities.

Let the state of an international system be:

$$Z_t = (Z_t^{economic}, Z_t^{political}, Z_t^{security}). \quad (76)$$

An external geopolitical shock can affect:

$$Z_t \rightarrow E_t \rightarrow K_t \rightarrow S_t \rightarrow Z_{t+1}. \quad (77)$$

This is a reflexive international system.

However, geopolitical interpretation must not be substituted for economic evidence.

A statistical cluster does not by itself establish a military alliance, and a currency correlation does not establish strategic alignment.

21 Diplomacy

Diplomacy can be represented as a mechanism for reducing uncertainty.

Let:

$$\Omega_t \quad (78)$$

denote the information set available to market participants.

Diplomatic communication may change:

$$\Omega_t \rightarrow \Omega_{t+1}. \quad (79)$$

This changes expectations:

$$E_t[P_{t+1} \mid \Omega_t] \rightarrow E_{t+1}[P_{t+2} \mid \Omega_{t+1}]. \quad (80)$$

Diplomacy can therefore enter the financial model without reducing international relations to finance.

22 Welfare Economics

The existence of an arbitrage opportunity does not automatically imply an increase in social welfare.

Let social welfare be:

$$W = \sum_{i=1}^N \omega_i U_i. \quad (81)$$

A private arbitrage profit is:

$$\Pi_i > 0. \quad (82)$$

It does not follow that:

$$\Delta W > 0. \quad (83)$$

The welfare effect depends upon liquidity provision, market efficiency, distribution, externalities, taxation, and resource allocation.

Reflexivity makes this distinction even more important because private actions may affect the prices faced by other agents.

23 Warfare Economics

The present paper does not treat economic models as predictions of war.
 Nevertheless, warfare economics illustrates the importance of feedback.
 Let national resources be:

$$R_t = R_t^{civilian} + R_t^{defense}. \quad (84)$$

A security shock can cause:

$$R_t^{defense} \uparrow \quad (85)$$

and therefore:

$$R_t^{civilian} \downarrow \quad (86)$$

under a fixed resource constraint.

The resulting economic consequences can alter subsequent strategic decisions.

Thus:

$$\text{security shock} \rightarrow \text{resource allocation} \rightarrow \text{economic conditions} \rightarrow \text{strategic expectations}. \quad (87)$$

The structure is again reflexive.

24 A Critical Distinction Between Model and Interpretation

The distinction between statistical classification and substantive interpretation is essential.

Suppose clustering produces:

$$C = \{C_1, C_2, \dots, C_K\}. \quad (88)$$

The clustering algorithm establishes a partition according to the chosen distance function and algorithm.

It does not automatically establish:

$$C_i = \text{political alliance}. \quad (89)$$

Nor:

$$C_i = \text{military coalition}. \quad (90)$$

Nor:

$$C_i = \text{causal economic structure}. \quad (91)$$

Those are additional hypotheses.

This distinction is especially important because the uploaded World War III paper moves from statistical clusters to geopolitical interpretations. Its clustering procedure produces singleton and larger clusters, after which substantive geopolitical conclusions are proposed.

The present paper therefore treats those conclusions as interpretations, not as mathematical consequences of clustering.

25 The Hungary Supermodel

The proposed model is:

$$X_{t+1} = F(X_t, E_t, P_t, K_t, G_t, \varepsilon_t), \quad (92)$$

where:

$$X_t = \text{economic state}, \quad (93)$$

$$E_t = \text{expectations}, \quad (94)$$

$$P_t = \text{policy}, \quad (95)$$

$$K_t = \text{capital flows}, \quad (96)$$

$$G_t = \text{geopolitical information}, \quad (97)$$

$$\varepsilon_t = \text{exogenous shocks}. \quad (98)$$

Expectations are endogenous:

$$E_t = H(X_t, P_t, G_t, E_{t-1}). \quad (99)$$

Policy is also potentially endogenous:

$$P_t = J(X_t, E_t, G_t). \quad (100)$$

Capital flows respond to expected returns:

$$K_t = K(E_t[R_{t+1}], \sigma_t, L_t). \quad (101)$$

The exchange rate then responds to capital flows:

$$S_{t+1} = S_t + \lambda K_t + \eta_t. \quad (102)$$

The resulting system is:

$$E_t \rightarrow K_t \rightarrow S_t \rightarrow P_t \rightarrow E_{t+1}. \quad (103)$$

This is the proposed Hungarian Reflexivity Supermodel.

26 Econometric Identification

A practical empirical specification can begin with:

$$\Delta S_t = \alpha + \beta_1 \Delta i_t + \beta_2 \Delta E_t + \beta_3 \Delta K_t + \beta_4 \Delta P_t + \beta_5 Z_t + \varepsilon_t. \quad (104)$$

To test reflexive feedback, one can estimate:

$$E_{t+1} = \gamma_0 + \gamma_1 S_t + \gamma_2 P_t + \gamma_3 K_t + \gamma_4 E_t + u_t. \quad (105)$$

A vector autoregression can be represented as:

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + \varepsilon_t. \quad (106)$$

For:

$$Y_t = \begin{bmatrix} S_t \\ E_t \\ K_t \\ P_t \\ \pi_t \\ Y_t^{GDP} \end{bmatrix}, \quad (107)$$

one can test whether innovations in expectations predict subsequent prices and whether price innovations subsequently predict expectations.

27 Granger-Style Feedback

Let X_t represent market prices and E_t expectations.

Test:

$$H_0 : \gamma_1 = \gamma_2 = \dots = \gamma_p = 0 \quad (108)$$

in an equation where E_t is explained by lagged X_t .

Similarly test whether lagged expectations predict prices.

Evidence of bidirectional predictive content would be consistent with a feedback mechanism, although it would not by itself prove structural causality.

This distinction is important:

$$\text{predictive causality} \neq \text{structural causality}. \quad (109)$$

28 State-Space Representation

A state-space model can represent latent expectations.

Let:

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad (110)$$

and:

$$y_t = Cx_t + v_t. \quad (111)$$

Here x_t can contain latent expectations, perceived risk, and market regime variables.

A Kalman-filter architecture can estimate latent states when the system is approximately linear.

For nonlinear systems, particle filtering or nonlinear state-space methods can be considered.

29 Machine Learning Architecture

A machine-learning system can complement, but should not replace, econometric identification.

Potential features include:

$$Z_t = [\Delta S_t, \Delta i_t, CIP_t, \sigma_t, K_t, \pi_t, Y_t, E_t]. \quad (112)$$

A supervised classifier can estimate:

$$P(A_t = 1 \mid Z_t), \quad (113)$$

where $A_t = 1$ represents an estimated arbitrage window.

A second model can estimate:

$$P(R_t = 1 \mid Z_t), \quad (114)$$

where $R_t = 1$ represents a reflexive regime.

The two models should not be conflated.

30 Machine-Learning Pseudocode

INPUT:

HUF spot prices
HUF forward prices
interest rates
inflation
volatility
liquidity measures
capital-flow data
policy announcements

geopolitical-event indicators

STEP 1:
clean and synchronize all time series

STEP 2:
calculate bilateral arbitrage measures

STEP 3:
calculate triangular arbitrage measures

STEP 4:
calculate CIP deviations

STEP 5:
subtract transaction costs

STEP 6:
construct lagged expectation and market variables

STEP 7:
estimate econometric feedback models

STEP 8:
train machine-learning regime classifier

STEP 9:
evaluate out-of-sample performance

STEP 10:
compare Arbitrage and Reflexivity models

OUTPUT:
arbitrage probability
reflexive-regime probability
estimated feedback coefficients
stability diagnostics
economic interpretation

31 Artificial Intelligence and Agent-Based Modelling

A multi-agent representation can contain:

$$\mathcal{A} = \{\text{households, firms, banks, investors, central bank, government, foreign investors}\}. \quad (115)$$

Each agent i possesses a state:

$$s_{i,t}. \quad (116)$$

The agent chooses:

$$a_{i,t} = \pi_i(s_{i,t}). \quad (117)$$

The environment then evolves:

$$S_{t+1} = F(S_t, a_{1,t}, \dots, a_{N,t}). \quad (118)$$

The policy functions themselves can be learned:

$$\pi_i^* = \arg \max_{\pi_i} \mathbb{E} \left[\sum_{t=0}^T \delta^t r_{i,t} \right]. \quad (119)$$

This architecture makes it possible to simulate reflexive feedback without assuming a representative agent.

32 Algorithm for Detecting Reflexive Regimes

1. Collect synchronized market, policy, macroeconomic, and event data.
2. Construct arbitrage measures.
3. Construct CIP deviations.
4. Estimate transaction-cost-adjusted profitability.
5. Estimate lagged feedback between prices and expectations.
6. Identify structural breaks.
7. Estimate latent market regimes.
8. Test whether feedback coefficients change across regimes.
9. Conduct out-of-sample validation.
10. Compare predictive performance against an arbitrage-only benchmark.
11. Test robustness to alternative specifications.
12. Evaluate whether model publication or model-generated signals alter subsequent observations.

33 Reflexivity as a Control Problem

The system can be represented as:

$$x_{t+1} = f(x_t, u_t). \quad (120)$$

The controller observes:

$$y_t = h(x_t). \quad (121)$$

In a conventional control problem, the system does not necessarily react to the controller's model.

In a reflexive system:

$$u_t = \pi(y_t, M_t), \quad (122)$$

where M_t is the model used by the controller.

If agents know M_t , then:

$$M_t \rightarrow u_t \rightarrow x_{t+1} \rightarrow y_{t+1}. \quad (123)$$

The model therefore becomes part of the controlled system.

This provides a computer-science interpretation of reflexivity.

34 The Model-Observation Feedback Problem

Suppose:

$$M_t = f(D_t). \quad (124)$$

If the model becomes public:

$$D_{t+1} = g(D_t, f(D_t)). \quad (125)$$

Therefore:

$$D_{t+1} \neq g(D_t) \quad (126)$$

in general.

The data-generating process has changed because the model was observed.

This creates a fundamental problem for empirical validation.

A model can change the environment against which its predictive accuracy is evaluated.

35 Proof of the Basic Reflexive Feedback Proposition

Proposition. Suppose market price depends on expectations and expectations depend on market price. Then the equilibrium dynamics cannot generally be represented as a one-way exogenous-price process.

Proof.

Assume:

$$P_{t+1} = f(E_t), \quad (127)$$

$$E_{t+1} = g(P_{t+1}). \quad (128)$$

Substitution yields:

$$E_{t+1} = g(f(E_t)). \quad (129)$$

Thus:

$$E_{t+1} = h(E_t), \quad (130)$$

where:

$$h = g \circ f. \quad (131)$$

The expectation variable therefore evolves according to a function generated by the interaction of expectations and prices.

Consequently, the system is dynamically recursive.

□

36 Arbitrage Embedded Within Reflexivity

The central synthesis of this paper is:

$$\boxed{\text{Reflexivity} \supset \text{Arbitrage Mechanisms}} \quad (132)$$

in the sense that an arbitrage opportunity may be a local phenomenon inside a larger reflexive system. Suppose:

$$A_t = f(P_t). \quad (133)$$

Arbitrageurs respond:

$$Q_t^{arb} = q(A_t). \quad (134)$$

Their trading changes prices:

$$P_{t+1} = P_t + \lambda Q_t^{arb}. \quad (135)$$

Then:

$$A_{t+1} = f(P_{t+1}). \quad (136)$$

The arbitrage mechanism itself becomes one component of the reflexive feedback system.

37 Hungary as a Candidate Reflexive Laboratory

The empirical objective should therefore be:

$$\text{Test Reflexivity against Arbitrage.} \quad (137)$$

The null model can be:

$$M_A : \text{ price discrepancies are primarily corrected by arbitrage.} \quad (138)$$

The alternative model is:

$$M_R : \text{ market outcomes and expectations recursively determine one another.} \quad (139)$$

The comparison should include:

Table 1: Candidate Model Comparison		
Dimension	Arbitrage	Reflexivity
Primary object	Price discrepancy	Feedback system
Main mechanism	Convergence	Recursive interaction
Expectations	Often secondary	Endogenous
Policy	Constraint or input	Potential feedback variable
Prices	Signals	Signals and state variables
Agents	Exploit discrepancies	Alter observed system
Equilibrium	Central concept	Potentially unstable/dynamic
Empirical tool	Cycle detection	Dynamic feedback estimation
Graph theory	Central	Useful but insufficient
Machine learning	Opportunity detection	Regime detection

38 Statistical Testing Strategy

A rigorous empirical study should report:

$$R^2, \text{ } RMSE, \text{ } MAE, \text{ } AIC, \text{ } BIC, \text{ } \text{ out-of-sample loss.} \quad (140)$$

However, model selection should not depend exclusively on predictive accuracy.

A reflexive model should additionally be tested for:

1. parameter stability,
2. structural breaks,
3. feedback direction,
4. regime dependence,
5. endogeneity,
6. omitted-variable sensitivity,

7. transaction-cost robustness,
8. latency sensitivity,
9. model-publication effects.

39 Identification Problems

The major econometric challenge is simultaneity.

Suppose:

$$P_t = \alpha E_t + \varepsilon_t, \quad (141)$$

$$E_t = \beta P_t + u_t. \quad (142)$$

Ordinary least squares can produce biased estimates because:

$$E_t \not\perp \varepsilon_t. \quad (143)$$

Potential approaches include:

1. instrumental variables,
2. structural vector autoregressions,
3. high-frequency identification,
4. event studies,
5. local projections,
6. state-space models,
7. synthetic controls where appropriate,
8. causal machine learning,
9. regime-switching models.

No single method should be treated as universally sufficient.

40 Structural Breaks

A reflexive system may exhibit structural changes.

Let:

$$\theta_t = \begin{cases} \theta_1, & t < \tau, \\ \theta_2, & t \geq \tau. \end{cases} \quad (144)$$

The change point τ may correspond to a major policy, institutional, financial, or geopolitical event. A model assuming:

$$\theta_1 = \theta_2 \quad (145)$$

may therefore incorrectly classify the system.

This provides another reason why a static arbitrage test cannot by itself identify the complete Hungarian economic structure.

41 Network Science

Hungary can be represented as a node in a broader financial network.

Let:

$$G_t = (V, E_t, W_t). \quad (146)$$

The weighted adjacency matrix is:

$$A_t = [a_{ij,t}]. \quad (147)$$

Centrality measures include:

$$C_i^{degree} = \sum_j a_{ij}, \quad (148)$$

$$C_i^{between} = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}}, \quad (149)$$

where σ_{st} denotes the number of shortest paths between s and t , and $\sigma_{st}(i)$ denotes those passing through i .

Reflexivity can alter network topology because agents change their connections in response to perceived risk.

42 Information Theory

Let uncertainty be represented by entropy:

$$H(X) = - \sum_x p(x) \log p(x). \quad (150)$$

Information arrivals change the distribution:

$$p_t(x) \rightarrow p_{t+1}(x). \quad (151)$$

Market participants then react to the information.

Thus:

$$\text{information} \rightarrow \text{belief} \rightarrow \text{trade} \rightarrow \text{price} \rightarrow \text{new information}. \quad (152)$$

Reflexivity can therefore be interpreted as an information-feedback process.

43 Game Theory

Let players choose actions:

$$a_i \in A_i. \quad (153)$$

A Nash equilibrium satisfies:

$$u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*) \quad (154)$$

for every feasible a_i .

In a reflexive financial system, however, information about the equilibrium itself may alter strategic behaviour.

The equilibrium is therefore potentially endogenous to beliefs about the equilibrium.

This creates a higher-order strategic problem.

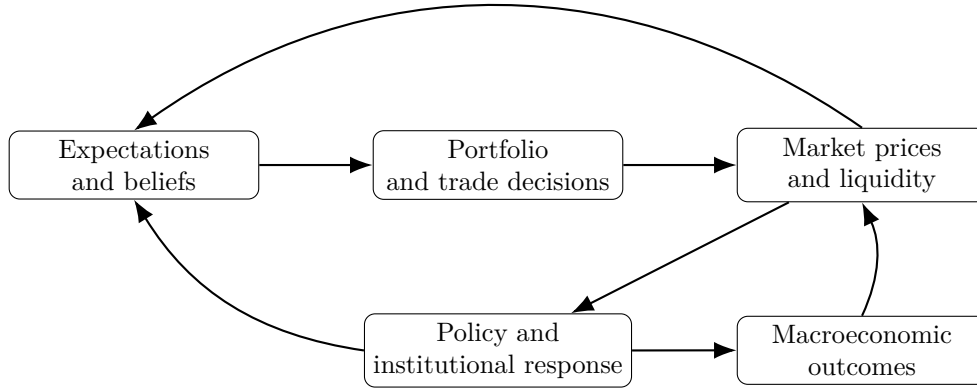


Figure 2: Proposed architecture.

44 International Political Economy

A national currency can be viewed simultaneously as:

$$\text{financial asset} + \text{policy instrument} + \text{national institution} + \text{international exchange medium}. \quad (155)$$

This makes a purely financial explanation incomplete whenever political institutions materially affect expectations.

The C4Hypermodel can therefore combine:

$$\text{Finance} + \text{Economics} + \text{Politics} + \text{Institutions}. \quad (156)$$

45 A Unified Reflexive Architecture

The proposed architecture is:

The architecture is intentionally multidirectional.

There is no requirement that causality operate along a single arrow.

46 The Seven-Supermodel Interpretation

Within the proposed C4Hypermodel architecture, the present paper treats Arbitrage as one supermodel and Reflexivity as another.

The precise complete taxonomy of all seven supermodels is not assumed here, because it is not necessary to establish the Hungary result.

The relevant distinction is:

$$\text{Arbitrage} = \text{one supermodel}, \quad (157)$$

while:

$$\text{Reflexivity} = \text{candidate supermodel for Hungary}. \quad (158)$$

The paper deliberately avoids inventing names for the remaining supermodels without empirical or theoretical justification.

47 The Soros Problem

The historical association between George Soros and Black Wednesday is useful because it illustrates the distinction between arbitrage and reflexivity.

A pure arbitrage interpretation would seek:

$$P_A - P_B > 0 \quad (159)$$

and exploit the discrepancy.

A reflexive interpretation asks:

$$\text{How do expectations about the sustainability of a policy regime} \quad (160)$$

affect:

$$\text{trading behaviour} \rightarrow \text{market pressure} \rightarrow \text{policy response} \rightarrow \text{expectations?} \quad (161)$$

The second question is structurally richer.

Historical analogy should not be confused with empirical proof that Hungary follows the same mechanism.

48 Model Risk

The C4Hypermodel itself creates model risk.

Let:

$$\widehat{M} = M + \epsilon_M. \quad (162)$$

If agents respond to $\widehat{M}$, then model error can propagate:

$$\epsilon_M \rightarrow \text{belief error} \rightarrow \text{portfolio adjustment} \rightarrow \text{price change}. \quad (163)$$

The resulting error is no longer purely statistical.

It becomes endogenous.

This is why testing a reflexive model can be risky.

49 Ethics

A model capable of influencing financial behaviour should distinguish:

$$\text{prediction} \neq \text{recommendation}. \quad (164)$$

Similarly:

$$\text{economic model} \neq \text{political directive}. \quad (165)$$

The purpose of this paper is explanatory and empirical.

It does not prescribe political action, military action, or financial conduct.

50 Limits of the Present Paper

The present paper does not claim to have established empirically that Reflexivity is the statistically dominant explanation of Hungarian financial dynamics.

The necessary Hungarian dataset has not yet been constructed in this paper.

The paper therefore makes a theoretical and methodological proposition:

$$\boxed{\text{Reflexivity is the candidate Hungarian supermodel.}} \quad (166)$$

The empirical test remains future work.

This distinction is essential for scientific integrity.

Table 2: Proposed Hungarian Reflexivity Dataset

Variable	Description
HUF spot rates	High-frequency exchange rates
HUF forward rates	Forward-market information
Policy rate	Monetary-policy stance
Inflation	Consumer-price dynamics
GDP	Real economic activity
Capital flows	Cross-border financial flows
Bond yields	Sovereign financing conditions
Volatility	Market uncertainty
Liquidity	Market depth and trading conditions
CIP deviation	Covered-interest-parity measure
Political events	Policy and institutional information
Geopolitical events	External strategic information
Expectations	Survey or market-implied expectations

51 Proposed Hungarian Dataset

A future empirical dataset should contain:

The data should be sampled at multiple frequencies where possible.

Daily data can provide macroeconomic context.

Intraday data can identify market microstructure.

Event-time data can identify immediate responses to information.

52 Research Design

The research program can be organized into four stages.

52.1 Stage One: Arbitrage Baseline

Estimate:

$$\Pi_t^{arb}. \quad (167)$$

Determine whether deviations survive transaction costs.

52.2 Stage Two: Reflexive Feedback

Estimate:

$$E_{t+1} = f(P_t, E_t, K_t, POL_t). \quad (168)$$

and:

$$P_{t+1} = g(E_t, K_t, POL_t). \quad (169)$$

52.3 Stage Three: Regime Detection

Estimate:

$$Pr(R_t = r \mid X_t). \quad (170)$$

for latent regimes r .

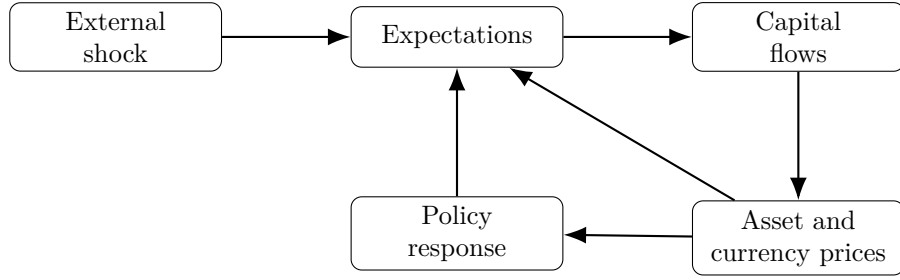


Figure 3: Simplified causal structure.

52.4 Stage Four: Out-of-Sample Validation

Compare:

$$L(M_A) \quad \text{and} \quad L(M_R), \quad (171)$$

where L is an out-of-sample loss function.

A model should not be declared superior solely because it fits the historical sample more closely.

53 Robustness Tests

The Hungarian Reflexivity Supermodel should survive:

1. alternative sampling frequencies;
2. alternative exchange-rate definitions;
3. alternative transaction-cost assumptions;
4. alternative lag structures;
5. alternative volatility estimators;
6. alternative expectations measures;
7. alternative event classifications;
8. placebo events;
9. rolling-window estimation;
10. expanding-window estimation;
11. out-of-sample validation.

54 Causal Diagram

A simplified causal structure is:

The feedback arrows are the defining feature.

55 A General Reflexivity Theorem

Theorem. If expectations affect economic outcomes and economic outcomes affect expectations, then the economic system contains a feedback channel that cannot be represented completely by a one-directional exogenous-expectation model.

Proof.

Suppose:

$$Y_{t+1} = F(E_t, X_t) \quad (172)$$

and:

$$E_{t+1} = G(Y_{t+1}, X_{t+1}). \quad (173)$$

Then:

$$E_{t+1} = G(F(E_t, X_t), X_{t+1}). \quad (174)$$

Thus future expectations depend on functions of current expectations.

The system is therefore recursive.

□

56 Interpretation

The theorem does not imply instability.

It does not imply irrationality.

It does not imply that all prices are arbitrary.

It does not imply that arbitrage disappears.

It implies only that the state of the system depends partly upon the expectations and actions of agents within the system.

This is sufficient to motivate Reflexivity as a candidate supermodel.

57 Conclusion

The central conclusion is deliberately narrow.

The evidence and mathematical machinery considered in this paper support the distinction between two mechanisms:

$$\text{Arbitrage} = \text{exploitation of price inconsistency}, \quad (175)$$

and:

$$\text{Reflexivity} = \text{feedback between expectations, actions, and outcomes}. \quad (176)$$

The uploaded currency-arbitrage studies demonstrate that geopolitical complexity does not automatically produce exploitable currency arbitrage. One study found no exploitable bilateral or triangular opportunities in its nine-currency snapshot, while another reported recurring transaction-cost-adjusted opportunities in Scandinavian currency markets. The contrast itself demonstrates the importance of market structure, frequency, liquidity, and institutional constraints.

The proposed Hungarian hypothesis is therefore:

$$\boxed{\text{Hungary} \rightarrow \text{Reflexivity Supermodel}} \quad (177)$$

with:

$$\boxed{\text{Arbitrage} \subset \text{the broader financial mechanism set.}} \quad (178)$$

The hypothesis remains to be empirically tested.

The appropriate scientific next step is not to force Hungary into the Arbitrage model.

It is to construct the Hungarian dataset and determine whether recursive feedback between expectations, markets, policy, institutions, and capital flows provides measurable explanatory power.

If it does, then Hungary provides a natural empirical case for a C4Hypermodel supermodel based on Reflexivity.

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Glossary

Arbitrage

The exploitation of price discrepancies across markets or instruments subject to the relevant execution and risk constraints.

Bid–Ask Spread

The difference between the price at which a market participant can sell and the price at which the participant can buy.

Capital Flow

A movement of financial capital between countries, markets, institutions, or asset classes.

CIP

Covered Interest Parity, the theoretical relationship connecting spot exchange rates, forward exchange rates, and interest-rate differentials.

C4Hypermodel

The higher-order modelling architecture within which multiple supermodels can represent distinct mechanisms of economic and social systems.

Endogeneity

A condition in which an explanatory variable is correlated with the disturbance term or is jointly determined with the dependent variable.

Expectations

Agents' beliefs concerning future economic, financial, political, or strategic states.

Feedback

A process in which the output of a system affects subsequent inputs.

Granger Predictive Causality

A statistical concept describing whether past values of one variable contain predictive information about another variable beyond its own history.

Market Microstructure

The study of trading mechanisms, liquidity, order flow, transaction costs, price formation, and market organization.

Negative Cycle

A directed cycle whose total logarithmic edge weight is negative, corresponding to an arbitrage opportunity in the currency-network representation.

Reflexivity

A recursive interaction in which participants' beliefs and actions affect the economic conditions that subsequently influence those beliefs and actions.

Regime

A relatively stable configuration of the statistical or economic properties of a dynamic system.

Supermodel

A higher-level model within the C4Hypermodel architecture representing a distinct economic or interdisciplinary mechanism.

Transaction Costs

The costs associated with executing an economic or financial transaction, including spreads, commissions, market impact, settlement costs, and latency-related costs.

Triangular Arbitrage

An arbitrage strategy involving three currencies and a closed sequence of currency conversions.

The End